



Inspiring Excellence

CSE370 : Database Systems
Project Report
Project Title : DonorDash

Group No : 03, CSE370 Lab Section : 03, Spring 2026		
ID	Name	Contribution
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24341017	S. M. Tahmidul Hoque	Clerk Operations & Predictive UI , Backend
24301078	Mohammad Ahonaf Bin Ameen	Auth & Public Portals , Backend

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Introduction

DonorDash is a centralized, highly specialized healthcare logistics platform designed to manage the complex allocation of critical and scarce medical resources, specifically blood and rare plasma. Rather than functioning as a standard patient-doctor portal this system operates entirely as a high stakes matching engine. It cross references patient needs with a massive network of donors and hospitals, ensuring that life saving resources are distributed rapidly, fairly and safely based on biological compatibility and medical urgency.

Initial Management Features:

1. **Role-Based Access Control:** Secure login portals that verify credentials and automatically route users (Hospital Admins, Medical Officers, Data Clerks) to their specialized, permission-based dashboards.
2. **Hybrid Registration Engine:** Frontend data entry interfaces that process new donor and patient records through both public self-registration forms and direct manual entry within the Data Clerk hub.
3. **Medical Approval Workflow:** An interactive dashboard queue where Medical Officers and Clerks evaluate pending resource requests and officially approve patients to "Active" status on the hospital's priority waitlist.

Along with these initial management features our platform consists of nine exceptional features that will help hospitals to collect blood effectively and also provide vitals to those who are in need.

Project Features

ID, Name	Features [3 per member]	
24301485, Taufiq Mustafizur Rahman	Ft 1	Automated Resource Rebalancer
	Ft 2	Donor Loyalty Analyzer
	Ft 3	Best Match Scoring System
24341017, S. M. Tahmidul Hoque	Ft 1	Stockout Predictor
	Ft 2	Emergency Nearby Resource Locator
	Ft 3	Rare Trait Priority Scanner
24301078, Mohammad Ahonaf Bin Ameen	Ft 1	Waitlist Priority Escalator
	Ft 2	Hospital Wastage Tracker
	Ft 3	Suspicious Activity Monitor

The features are explained below:

1. Stockout Predictor

- Predicts exactly how many days are left until a hospital runs out of a specific blood group.

2. Automated Resource Rebalancer

- Automatically matches hospitals running out of blood with neighboring hospitals that have too much.

3. Donor Loyalty Analyzer

- Tracks what percentage of donors return to donate again the following year.
-

4. Hospital Wastage Tracker

- Calculates the percentage of blood units that expire before they can be used at each hospital.

5. Suspicious Activity Monitor

- Flags staff members who are registering an unusually high number of "Urgent" patients compared to average.

6. Rare Trait Priority Scanner

- Automatically flags patients who need a genetic match that exists in less than 1% of the donor database.

7. Emergency Nearby Resource Locator

- Instantly finds the three closest hospitals in nearby zones that have a required resource ready to go.

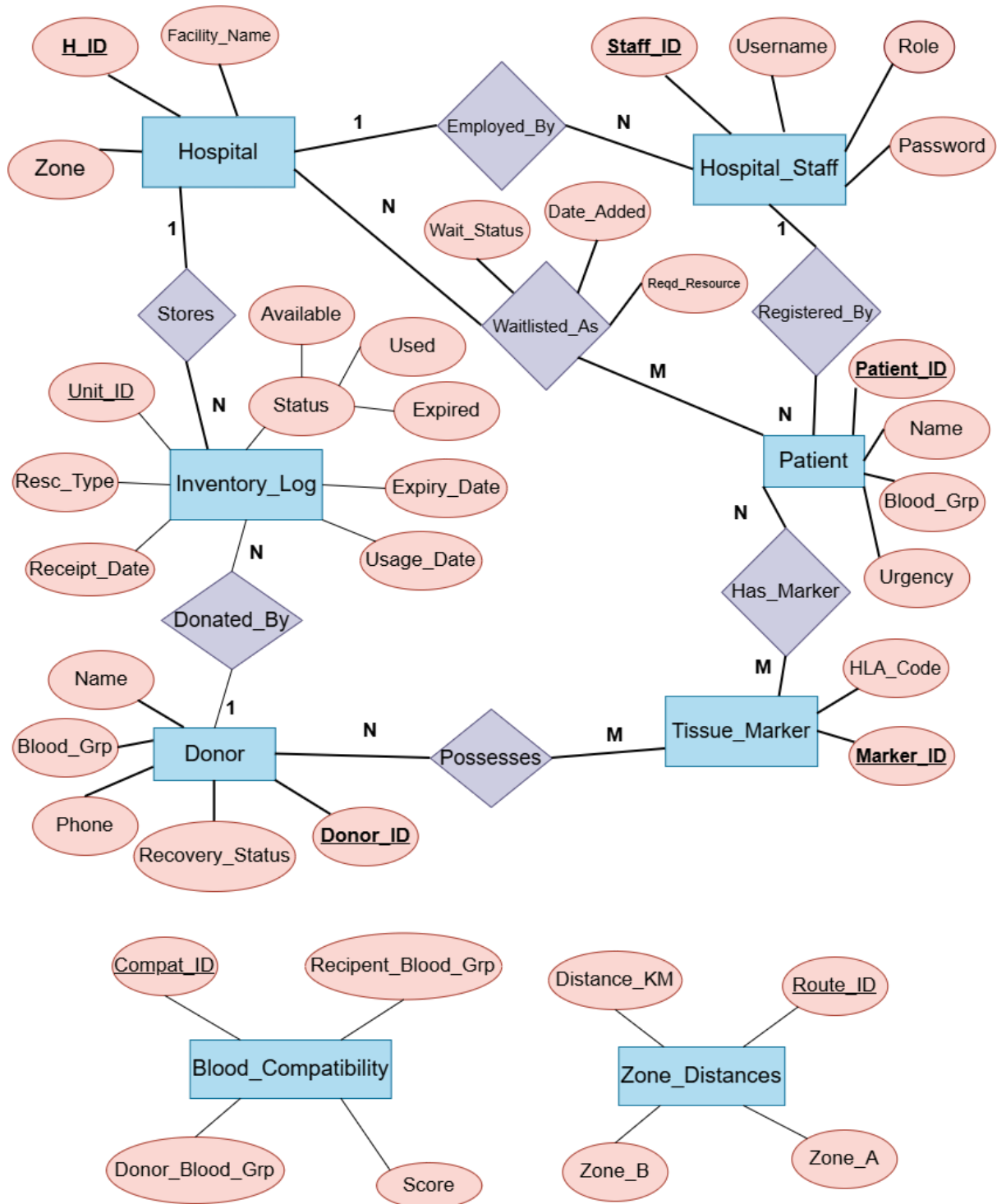
8. Waitlist Priority Escalator

- Bumps up the urgency of patients who have been waiting longer than 90% of historically treated patients.

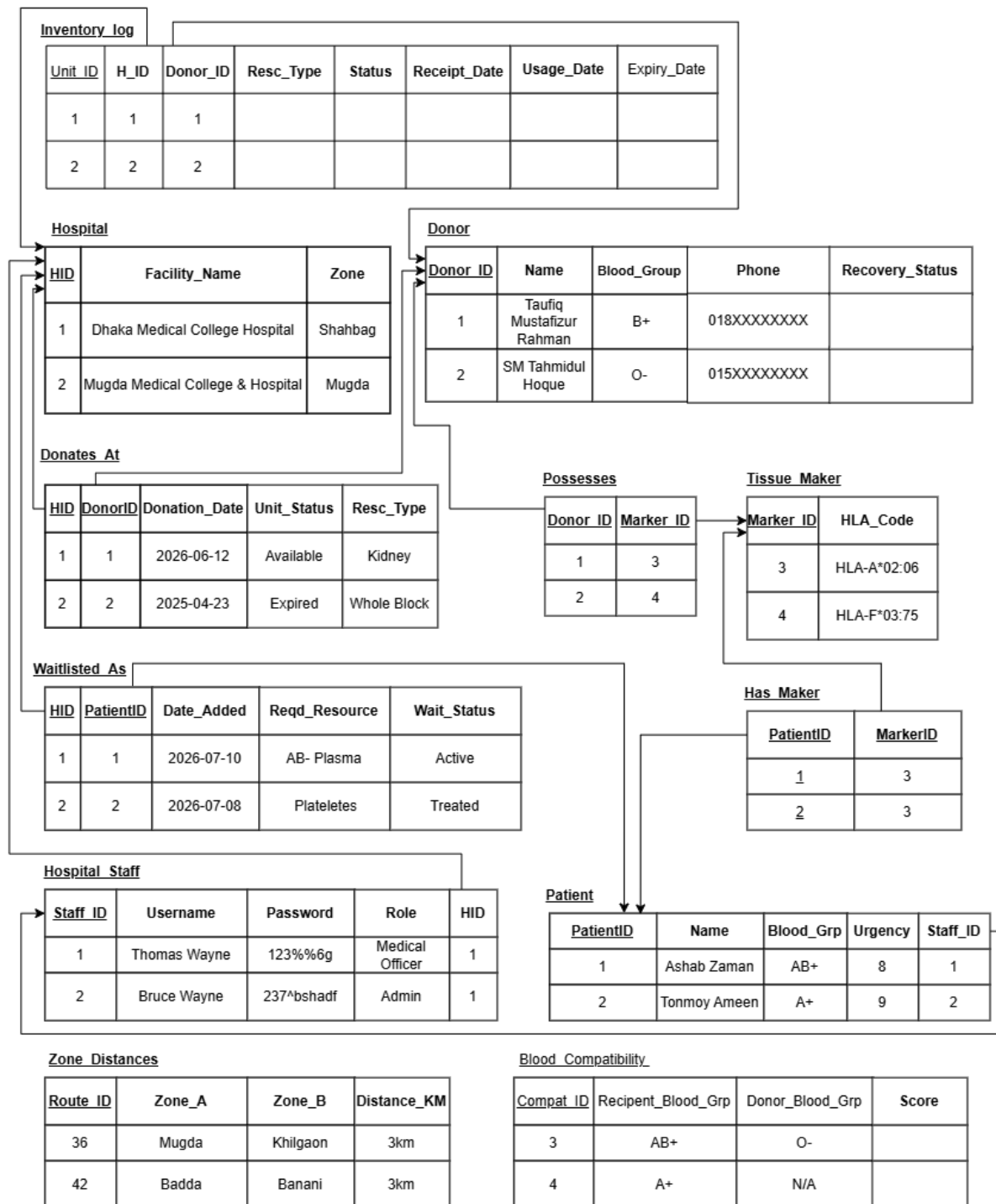
9. Best Match Scoring System

- Gives a numbered score to potential donors based on how perfectly their blood and tissue match the patient.

ER/EER Diagram



Schema Diagram



Normalization

- a. Explain if your converted Schema is in 1NF or not. If not, decompose it to 1NF.**

Ans: The converted Schema is in 1NF because there is no composite/ multivalued attributes.

- b. Explain if your converted Schema is in 2NF or not. If not, decompose it to 2NF.**

Can there be any partial functional dependencies in your relational schema?

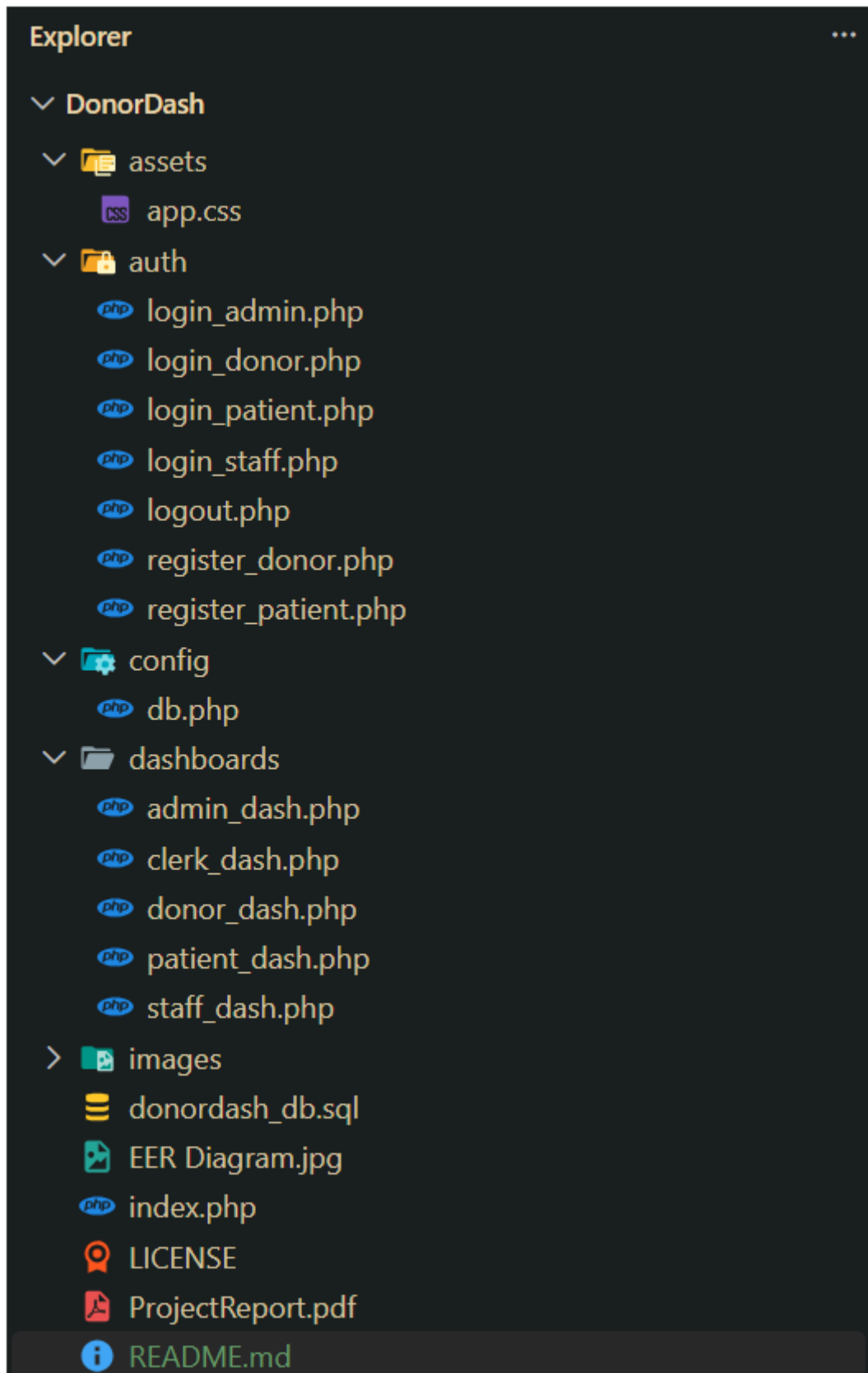
Ans: The converted Schema is in 2NF. There cannot be any partial functional dependencies so the schema is in 2NF.

- c. Explain if your converted Schema is in 3NF or not. If not, decompose it to 3NF.**

Can there be any transitive dependencies in your relational schema?

Ans: It is already in 3NF. There are no transitive dependencies in our relational schema.

Directory Structure

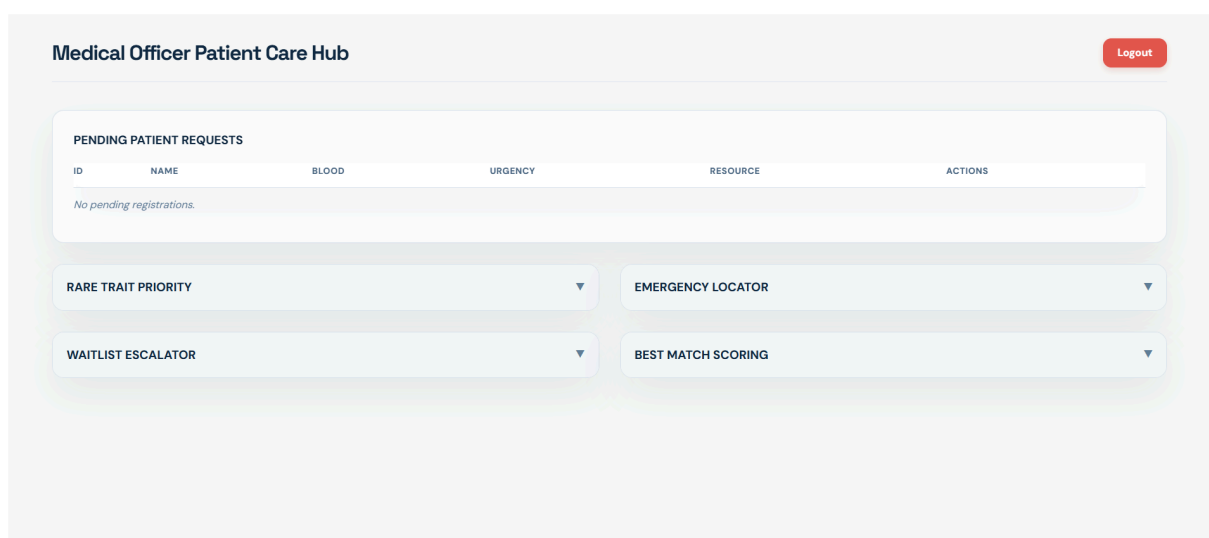
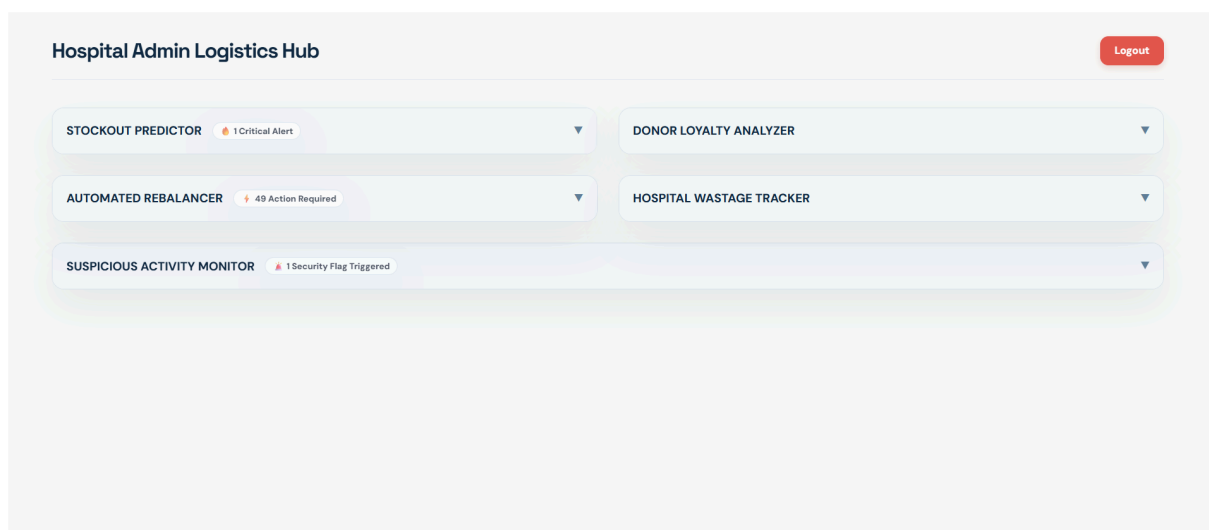


Frontend Development

The frontend was built with HTML, CSS and JavaScript integrated directly into PHP files to provide a responsive and user-friendly healthcare logistics interface. It includes authentication screens, dedicated role-based dashboards (Admin, Staff, Clerk, Donor, Patient) and specialized panels for resource tracking. User actions connect to the backend to dynamically update waitlists, map out nearby resources and display real time inventory stats.

Contribution of ID: 24301485, Name: Taufiq Mustafizur Rahman (Staff Operations & Matching UI)

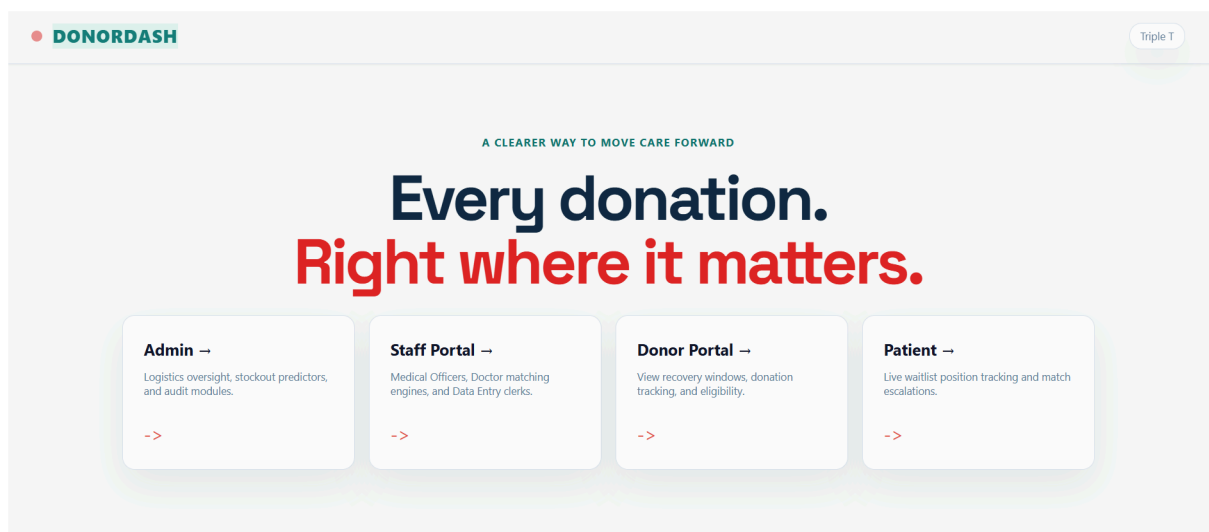
- Dashboards: Developed the Staff Dashboard ([staff_dash.php](#)) and Admin Dashboard ([admin_dash.php](#)).



- Features: Designed the UI for the Automated Resource Rebalancer, the Donor Loyalty Analyzer visualization and the Best Match Scoring System panel.

Contribution of ID: 24341017, Name: S. M. Tahmidul Hoque (Clerk Operations & Predictive UI)

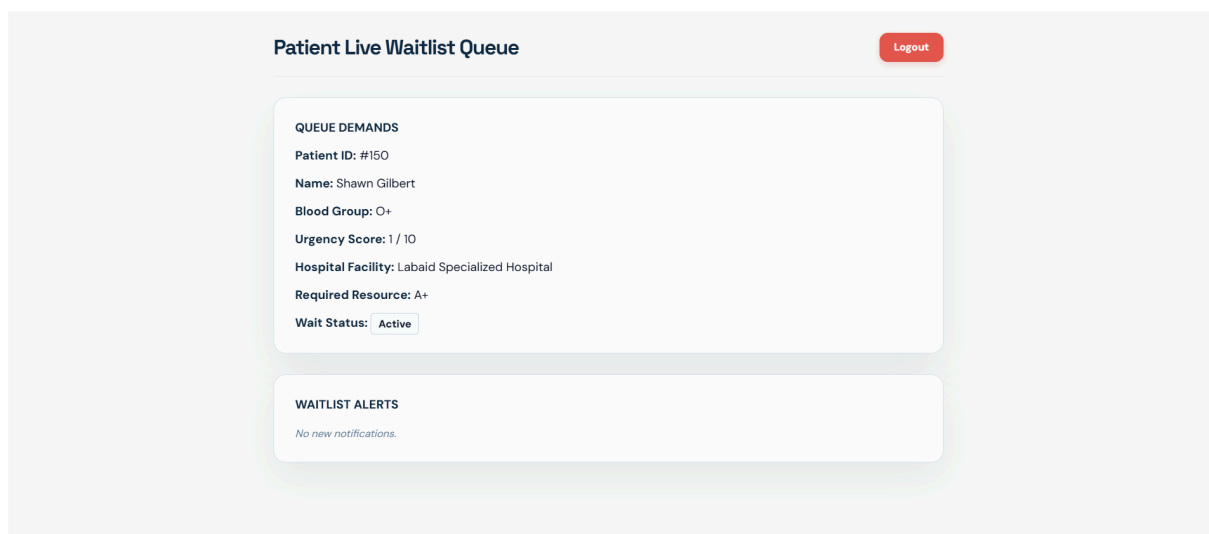
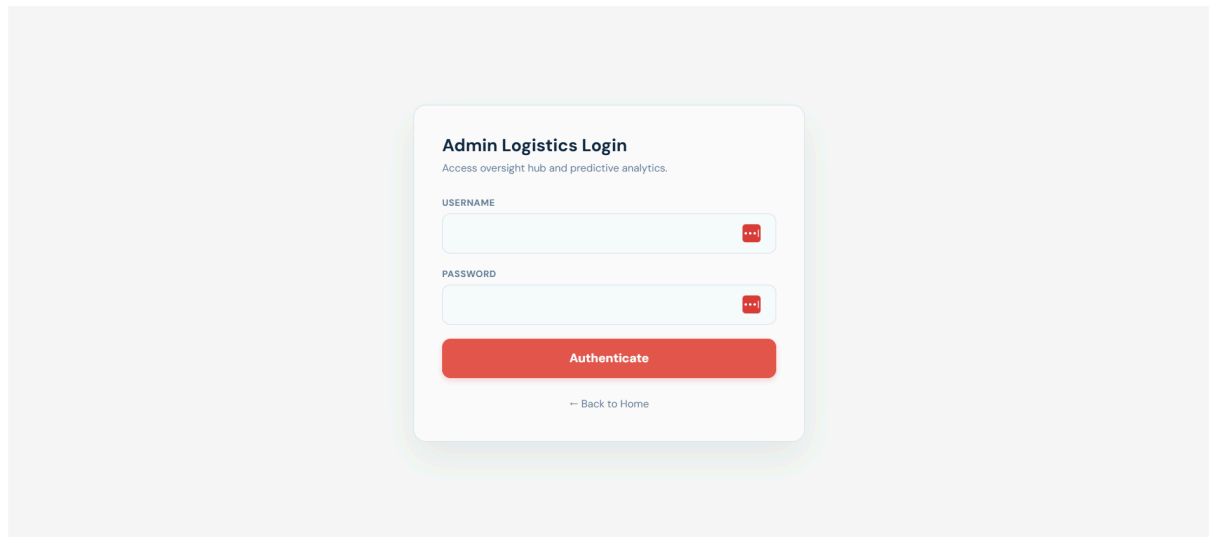
- Dashboards: Developed the Clerk Dashboard ([clerk_dash.php](#)) and the global stylesheet ([assets/app.css](#)) and index([index.php](#))

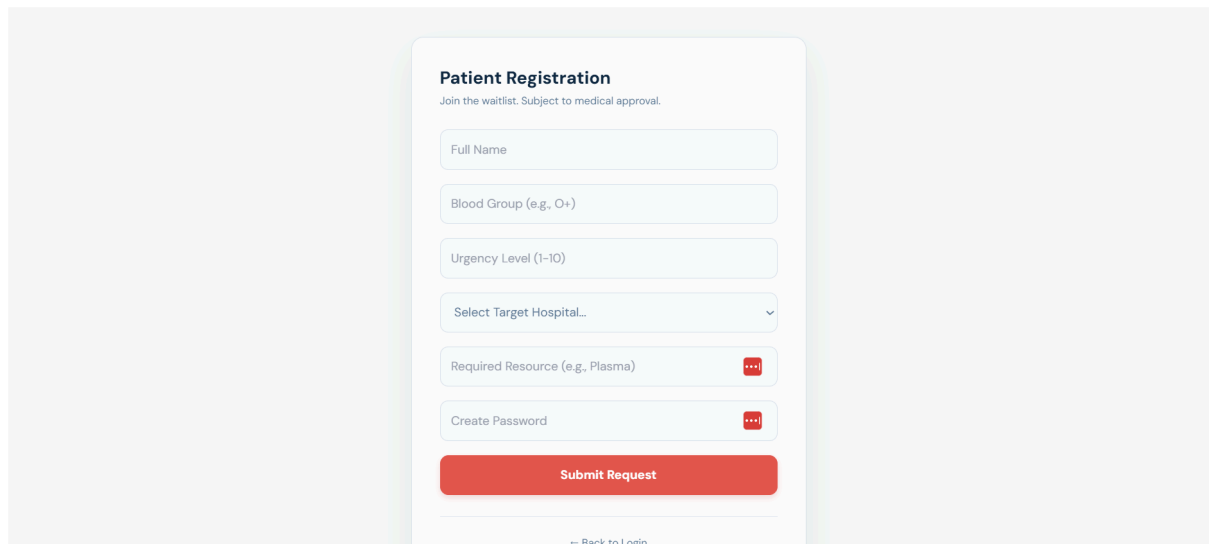


- Features: Built the interface for the Stockout Predictor, the interactive map views for the Emergency Nearby Resource Locator and the alert components for the Rare Trait Priority Scanner

Contribution of ID: 24301078, Name: Mohammad Ahonaf Bin Ameen (Auth & Public Portals)

- Dashboards & Auth: Developed the Patient and Donor Dashboards ([patient_dash.php](#), [donor_dash.php](#)) and the entire authentication flow including [auth/login_admin.php](#), [auth/login_donor.php](#), and [auth/register_patient.php](#).





The image shows a mobile app UI for a 'Patient Registration' form. The form is centered on a light gray background. It has a title 'Patient Registration' and a subtitle 'Join the waitlist. Subject to medical approval.' Below these are six input fields: 'Full Name', 'Blood Group (e.g., O+)', 'Urgency Level (1-10)', 'Select Target Hospital...' (a dropdown menu), 'Required Resource (e.g., Plasma)' (with a red three-dot icon), and 'Create Password' (with a red three-dot icon). At the bottom of the form is a red 'Submit Request' button. Below the button is a link that says '← Back to Login'.

Patient Registration
Join the waitlist. Subject to medical approval.

Full Name

Blood Group (e.g., O+)

Urgency Level (1-10)

Select Target Hospital...

Required Resource (e.g., Plasma)

Create Password

Submit Request

[← Back to Login](#)

- **Features:** Designed the UI for the Waitlist Priority Escalator, the Hospital Wastage Tracker charts and the Suspicious Activity Monitor logs.

Backend Development

The backend of this project was developed using PHP and MySQL to manage the complex allocation of critical and scarce medical resources. It handles role-based authentication, active waitlist management, dynamic inventory tracking and biological compatibility filtering. SQL queries execute the matching engine logic, delivering structured data back to the frontend to ensure resources are distributed rapidly and fairly.

Contribution of ID: 24301485, Name: Taufiq Mustafizur Rahman

1. Automated Resource Rebalancer

```
WITH AllNeeds AS (
    SELECT H_ID, Reqd_Resource AS Resource FROM Waitlisted_As WHERE
    Wait_Status = 'Active'
    UNION
    SELECT H_ID, Resc_Type AS Resource FROM Inventory_Log WHERE Status =
    'Available'
),
Demand AS (
    SELECT H_ID, Reqd_Resource, COUNT(Patient_ID) AS Demand_Count
    FROM Waitlisted_As WHERE Wait_Status = 'Active' GROUP BY H_ID,
    Reqd_Resource
),
Supply AS (
    SELECT H_ID, Resc_Type, COUNT(Unit_ID) AS Supply_Count
    FROM Inventory_Log WHERE Status = 'Available' GROUP BY H_ID,
    Resc_Type
),
Ratios AS (
    SELECT a.H_ID, a.Resource,
        CASE WHEN COALESCE(d.Demand_Count, 0) = 0 THEN 999.0
        ELSE COALESCE(s.Supply_Count, 0) * 1.0 / d.Demand_Count END AS
    Ratio
    FROM AllNeeds a
    LEFT JOIN Demand d ON a.H_ID = d.H_ID AND a.Resource =
    d.Reqd_Resource
    LEFT JOIN Supply s ON a.H_ID = s.H_ID AND a.Resource = s.Resc_Type
)
SELECT Deficit.H_ID AS Receiver_Hospital, Surplus.H_ID AS
Sender_Hospital, Deficit.Resource AS Reqd_Resource
FROM Ratios Deficit
JOIN Ratios Surplus ON Deficit.Resource = Surplus.Resource AND
Deficit.H_ID != Surplus.H_ID
WHERE Deficit.Ratio < 0.5 AND Surplus.Ratio > 2.0;
```

2. Donor Loyalty Analyzer

```
WITH FirstDonation AS (  
    SELECT  
        Donor_ID,  
        MIN(EXTRACT(YEAR FROM Receipt_Date)) AS Cohort_Year  
    FROM Inventory_Log  
    GROUP BY Donor_ID  
),  
SubsequentDonation AS (  
    SELECT DISTINCT  
        d.Donor_ID,  
        f.Cohort_Year  
    FROM Inventory_Log d  
    JOIN FirstDonation f ON d.Donor_ID = f.Donor_ID  
    WHERE EXTRACT(YEAR FROM d.Receipt_Date) = f.Cohort_Year + 1  
)  
SELECT  
    f.Cohort_Year,  
    COUNT(DISTINCT f.Donor_ID) AS Original_Cohort_Size,  
    COUNT(DISTINCT s.Donor_ID) AS Retained_Next_Year,  
    (COUNT(DISTINCT s.Donor_ID) * 100.0 / COUNT(DISTINCT  
f.Donor_ID)) AS Retention_Rate  
FROM FirstDonation f  
LEFT JOIN SubsequentDonation s ON f.Donor_ID = s.Donor_ID  
GROUP BY f.Cohort_Year;
```

3. Best Match Scoring System

```
WITH PatientInfo AS (  
    SELECT  
        Patient_ID,  
        Blood_Grp  
    FROM Patient  
    WHERE Patient_ID = $selected_patient_id  
),  
BloodScoring AS (  
    SELECT  
        d.Donor_ID,  
        d.Name AS Donor_Name,  
        d.Blood_Grp AS Donor_Blood,  
        bc.Score AS Blood_Score  
    FROM Donor d  
    CROSS JOIN PatientInfo p  
    JOIN Blood_Compatibility bc ON p.Blood_Grp =
```

```

bc.Recipient_Blood_Grp AND d.Blood_Grp = bc.Donor_Blood_Grp
  WHERE d.Recovery_Status = 'Recovered'
),
TissueScoring AS (
  SELECT
    dp.Donor_ID,
    (COUNT(ph.Marker_ID) * 10) AS Tissue_Score
  FROM Possesses dp
  JOIN Has_Marker ph ON dp.Marker_ID = ph.Marker_ID
  WHERE ph.Patient_ID = $selected_patient_id
  GROUP BY dp.Donor_ID
)
SELECT
  b.Donor_ID,
  b.Donor_Name,
  b.Donor_Blood,
  b.Blood_Score,
  COALESCE(t.Tissue_Score, 0) AS Tissue_Score,
  (b.Blood_Score + COALESCE(t.Tissue_Score, 0)) AS
Total_Match_Score
FROM BloodScoring b
LEFT JOIN TissueScoring t ON b.Donor_ID = t.Donor_ID
ORDER BY Total_Match_Score DESC, b.Blood_Score DESC;

```

Contribution of ID: 24341017, Name: S. M. Tahmidul Hoque

(Also responsible for *config/db.php* database connection architecture)

1. Stockout Predictor

```

WITH DailyUsage AS (
  SELECT
    H_ID,
    Resc_Type,
    COUNT(Unit_ID) / 30.0 AS Daily_Burn_Rate
  FROM Inventory_Log
  WHERE Status = 'Used' AND Usage_Date >= CURRENT_DATE - INTERVAL
30 DAY
  GROUP BY H_ID, Resc_Type
),
CurrentStock AS (
  SELECT
    H_ID,
    Resc_Type,
    COUNT(Unit_ID) AS Available_Units
  FROM Inventory_Log

```



```

        WHERE Status = 'Available'
        GROUP BY H_ID, Resc_Type
    )
    SELECT
        u.H_ID,
        u.Resc_Type,
        COALESCE(c.Available_Units, 0) AS Available_Units,
        ROUND(COALESCE(c.Available_Units, 0) / u.Daily_Burn_Rate, 1) AS
Days_Until_Depletion
    FROM DailyUsage u
    LEFT JOIN CurrentStock c ON u.H_ID = c.H_ID AND u.Resc_Type =
c.Resc_Type
    WHERE (COALESCE(c.Available_Units, 0) / u.Daily_Burn_Rate) <= 15.0
        AND u.H_ID = $f1_filter
    ORDER BY Days_Until_Depletion ASC;

```

2. Emergency Nearby Resource Locator

```

    SELECT
        TargetHospital.H_ID AS Sending_Hospital,
        TargetHospital.Facility_Name,
        zd.Distance_KM,
        i.Resc_Type,
        COUNT(i.Unit_ID) AS Available_Units
    FROM Zone_Distances zd
    JOIN Hospital TargetHospital ON zd.Zone_B = TargetHospital.Zone
    JOIN Inventory_Log i ON TargetHospital.H_ID = i.H_ID
    WHERE i.Status = 'Available'
    GROUP BY
        TargetHospital.H_ID,
        TargetHospital.Facility_Name,
        zd.Distance_KM,
        i.Resc_Type
    ORDER BY zd.Distance_KM ASC
    LIMIT 3;

```

3. Rare Trait Priority Scanner

```

    WITH TotalDonors AS (
        SELECT COUNT(DISTINCT Donor_ID) AS Total
        FROM Donor
    )

```

```

),
MarkerFrequency AS (
    SELECT
        Marker_ID,
        COUNT(Donor_ID) AS Donor_Count,
        (COUNT(Donor_ID) * 100.0 / (SELECT Total FROM TotalDonors))
    AS Rarity_Percent
    FROM Possesses
    GROUP BY Marker_ID
)
SELECT
    p.Patient_ID,
    p.Name,
    mf.Marker_ID,
    mf.Rarity_Percent
FROM Patient p
JOIN Has_Marker hm ON p.Patient_ID = hm.Patient_ID
JOIN MarkerFrequency mf ON hm.Marker_ID = mf.Marker_ID
WHERE mf.Rarity_Percent < 1.0;

```

Contribution of ID: 24301078, Name: Mohammad Ahonaf Bin Ameen

1. Waitlist Priority Escalator

```

SELECT
    Patient_ID,
    Reqd_Resource,
    Date_Added,
    CURRENT_DATE - Date_Added AS Days_Waiting,
    PERCENT_RANK() OVER (PARTITION BY Reqd_Resource ORDER BY
CURRENT_DATE - Date_Added ASC) AS Wait_Time_Percentile
FROM Waitlisted_As
WHERE Wait_Status = 'Active';

```

2. Hospital Wastage Tracker

```

SELECT
    H_ID,
    COUNT(Unit_ID) AS Total_Units_Received,

```

```

        SUM(CASE WHEN Status = 'Expired' THEN 1 ELSE 0 END) AS
Expired_Units,
        SUM(CASE WHEN Status = 'Used' THEN 1 ELSE 0 END) AS
Successfully_Used,
        ROUND(SUM(CASE WHEN Status = 'Expired' THEN 1 ELSE 0 END) *
100.0 / COUNT(Unit_ID), 2) AS Wastage_Percentage
FROM Inventory_Log
WHERE Receipt_Date >= CURRENT_DATE - INTERVAL 1 YEAR
GROUP BY H_ID
ORDER BY Wastage_Percentage DESC;

```

3. Suspicious Activity Monitor

```

WITH ClerkStats AS (
    SELECT
        hs.Staff_ID,
        COUNT(p.Patient_ID) AS Urgent_Registrations
    FROM Hospital_Staff hs
    JOIN Patient p ON hs.Staff_ID = p.Staff_ID
    WHERE p.Urgency >= 8 AND p.Registration_Date >= CURRENT_DATE -
INTERVAL 30 DAY
    GROUP BY hs.Staff_ID
),
GlobalAverages AS (
    SELECT
        AVG(Urgent_Registrations) AS Avg_Reg,
        STDDEV(Urgent_Registrations) AS Std_Dev
    FROM ClerkStats
)
SELECT
    c.Staff_ID,
    c.Urgent_Registrations
FROM ClerkStats c
CROSS JOIN GlobalAverages g
WHERE c.Urgent_Registrations > (COALESCE(g.Avg_Reg, 0) + (2 *
COALESCE(g.Std_Dev, 0)));

```

Source Code Repository -
<https://github.com/taufiqmrahman/DonorDash>

Conclusion

The project successfully progressed from initial conceptual design including comprehensive ER diagrams, schema mapping and strict 3NF normalization to a fully structured application architecture. By combining a PHP and MySQL backend with a responsive HTML, CSS and JavaScript frontend we successfully implemented nine advanced features, including predictive stockout tracking, automated rebalancing and complex biological matching algorithms. Ultimately, this collaborative effort, documented from the raw database schema to the final GitHub repository, demonstrates the successful practical application of advanced database engineering principles into a fully functional platform.

References

1. **UI/UX Design Inspiration:** Dribbble. "Skymetrics landing page". Utilized as the primary visual and structural reference for the frontend dashboard interfaces. Available at: <https://dribbble.com/shots/27682007-Skymetrics-landing-page>
2. **Frontend Styling Framework:** Tailwind Labs. "Tailwind CSS Documentation". Utilized for rapid UI development, grid layouts, and responsive dashboard design via CDN. Available at: <https://tailwindcss.com/docs>
3. **Database Management System:** Oracle. "MySQL 8.0 Reference Manual". Utilized for advanced query construction, including Common Table Expressions (CTEs), Aggregate Functions (`STDDEV`), and Window Functions (`PERCENT_RANK()`). Available at: <https://dev.mysql.com/doc/refman/8.0/en/>
4. **Backend Architecture & Security:** The PHP Group. "PHP: Hypertext Preprocessor Manual". Utilized for session management, MySQLi database connections, and secure password hashing (`password_verify`). Available at: <https://www.php.net/docs.php>
5. **Academic Foundation & Database Theory:** Elmasri, R., & Navathe, S. B. (2015). *Fundamentals of Database Systems* (7th ed.). Pearson. Utilized as the foundational text for designing the Entity-Relationship (ER) model, schema mapping, and ensuring 1NF, 2NF, and 3NF normalization compliance.